

Jem pays money into a savings scheme, A , over a period of 300 months.

Jem pays £20 into scheme A in month 1, £20.50 in month 2, £21 in month 3 and so on, so that the amounts Jem pays each month form an arithmetic sequence.

- (a) Show that Jem pays £69.50 into scheme A in month 100 (1)
- (b) Find the **total** amount that Jem pays into scheme A over the period of 300 months. (2)

Kim pays money into a different savings scheme, B , over the same period of 300 months.

In a model, the amounts Kim pays into scheme B increase by the same percentage each month, so that the amounts Kim pays each month form a geometric sequence.

Given that Kim pays

- £20 into scheme B in month 1
 - £250 into scheme B in month 300
- (c) use the model to calculate, to the nearest £10, the difference between the total amount paid into scheme A and the total amount paid into scheme B over the period of 300 months. (3)